

WACE Mathematics Methods · Cheat Sheet (Patterns + Notes)

General rules: show that: write every step · optimisation questions that require using calculus must verify the extremum (f'' or sign table, 1 mark) · calculator in radians (if $\pi/180$ appears when differentiating sin, it has been set to degrees) · include units in application questions · explanations must clearly state "which quantity + when + how much it changes + units", in context · probabilities to at least 4 decimal places (if the question specifies 3 decimal places, exact, or integer, follow that) · for CA questions write the equation solved and key output, not calculator syntax · read the question carefully for "first / initially" ($t = 0$), "more than" ($\geq k+1$), "at least"

1 Differentiation and its applications

① Direct differentiation (CF opening 2–4 marks)

- First identify the structure: product $\rightarrow$ product rule; fraction $\rightarrow$ quotient rule; composite $\rightarrow$ chain rule, often combined (e.g. $x^2 \ln(4x + 3)$).
- Standard derivatives: $(e^{kx})' = ke^{kx}$; $(\ln f)' = f'/f$; $(\sin kx)' = k \cos kx$; $(\tan x)' = 1/\cos^2 x$ (2017A-Q10 requires proof using the quotient rule).
- Δ Method marks require writing the two-term structure and at least one term correct; for $(2x^3+1)^5$ do not forget to multiply by $6x^2$; when differentiating e^{-x} do not miss the negative sign; if it says "do not simplify", you do not need to simplify.

② Differentiation from a table / graph of values

- First write the general formula (1 mark): $(fg)' = f'g + fg'$; $(g/h)' = (g'h - gh')/h^2$; $[h(g(x))]' = h'(g(x)) \cdot g'(x)$.
- Δ For composites, evaluate the inner function first: $h'(g(1)) = h'(3)$, not $h'(1)$ (2019 report: poor responses in composite differentiation).
- Graph: the gradient of a straight-line segment is the derivative; at a corner (sharp point) it is not differentiable.

③ "rate of change of $f'(x)$ " / interpreting the derivative

- This means find $f''(a)$ (recognising this is worth 1 mark on its own).
- Explanation template: "When $x = a$, the gradient of f is increasing at a rate of ... per unit of x "; $f'(a) = 0$ and $f''(a) < 0 \Rightarrow$ local maximum.

④ Full function analysis (10–15 mark extended question)

- Factorise f' ; write $e^x > 0$ (never zero) $\rightarrow$ solve $f' = 0 \rightarrow$ substitute back into the original function to find the y-coordinate (marked separately).
- Nature: $f''(a) > 0$ minimum, < 0 maximum; Δ when $f''(a) = 0$ you must use a sign table instead (2023 report).
- Points of inflection: $f'' = 0$ and f'' changes sign on both sides. $\Delta f' = f'' = 0$ alone is not enough (x^4 has a minimum at 0).
- Sketching: label coordinates of intercepts, stationary points and points of inflection; write equations of asymptotes (for $\ln(x+3)$, domain $x > -3$ and vertical asymptote are worth 1 mark separately); use open/closed circles at endpoints; concavity matches the sign of f'' .
- Example: 2021A-Q12 total 15 marks = differentiation 2 + stationary points and nature 7 + points of inflection 2 + sketching 4.

⑤ Finding coefficients from conditions / sketching from conditions

- Passes through a point $f(a) = b$; turning point $f'(a) = 0$; point of inflection $f''(a) = 0$; area $\int f = A$. First write the general forms of f' , f'' .
- Example: 2018A-Q15 "growth pauses at $t = 6$, then continues increasing" $\Rightarrow$ horizontal point of inflection $\Rightarrow P'(6) = 0$ and $P''(6) = 0$.
- Sketching from conditions (2022F-Q5): $f''(-1) > 0 \Rightarrow$ minimum; only two stationary points and $f'(2) > 0 \Rightarrow x = 1$ is a horizontal point of inflection.

⑥ Optimisation ★

- show that: use the constraint (fixed volume, perimeter 500 m, Pythagoras relationship) to eliminate one variable, substitute into the objective function, and simplify to the form given in the question.
- Set derivative = 0, reject unreasonable solutions ($r = 0$, θ out of range) $\rightarrow$ verify: calculate $A''(x_0) < 0$ at the stationary point, or draw a sign table $\rightarrow$ answer with all dimensions (both radius and height), round as required and include units.
- If the question says "You do not need to verify", you may omit verification; if only integer values are allowed, compare the values of the two adjacent integers (2020A-Q13: 74 and 75).

⑦ Increment formula $\delta y \approx f'(a) \cdot \delta x$

- Write δx ($\Delta 1^\circ = \pi/180$ rad; 1 month = $1/12$ year) $\rightarrow$ differentiate at the starting point $\rightarrow$ multiply; if asked for the new value, add $f(a)$.
- Example: $\ln 2.02 \approx \ln 2 + (1/2)(0.02) = 0.703$ (2021F-Q3).

⑧ Related rates of change / tangents

- $dy/dt = (dy/d\theta)(d\theta/dt)$: first use the geometric relationship to write the function (lighthouse $y = 12 \tan \theta$) $\rightarrow$ differentiate $\rightarrow$ substitute the values at that time (first use $y = 5$ to find $\cos \theta = 12/13$).
- Tangents: the point of contact satisfies $f'(a) =$ gradient, $f(a) =$ value on the tangent; tangent equation $y - f(a) = f'(a)(x - a)$.

⑨ Describing and interpreting (1–3 marks each year)

- Piecewise description: "For ... the company makes a loss; profit increases to a maximum of \$... at $x = ...$; then decreases."
- The maximum rate of increase is where the graph has its greatest gradient, i.e. at the point of inflection (2025A-Q9(c)).

2 Integration and its applications

① Antiderivatives and finding f from f'

- $\int e^{kx} = e^{kx}/k$; $\int \cos kx = \sin kx/k$; $\int (ax + b)^n = (ax + b)^{n+1}/[a(n+1)]$ ($n \neq -1$); $\int f'/f = \ln f(x) + c$ (in exams usually $f > 0$).
- Adjust the numerator to become the derivative of the denominator: $\int (3x+1)/(3x^2+2x+1) = \frac{1}{2} \ln(...)$ (adjusting the coefficient is worth 1 mark separately).
- Finding f from f' : integrate and write $+ c \rightarrow$ substitute the known point $\rightarrow$ write the full $f(x)$, simplifying using log laws if needed.

② "Hence" integration (common in CF)

- Fixed three steps: integrate both sides $\rightarrow$ the left side cancels using FTC $\rightarrow$ split the right side and rearrange.
- Example: $d/dx(x \ln x) = \ln x + 1 \Rightarrow \int \ln x \, dx = x \ln x - x + c$. This is often followed by a question asking for

expectation, a definite integral or area.

③ Definite integrals on graphs

- Split the region into rectangles, triangles and sectors; above the x-axis is positive, below is negative. $\int (f - 2) = \int f - 2 \times \text{interval length}$; $\int_a^b f' = f(b) - f(a)$ (read the two values directly from the graph).
- Find α such that $\int_0^\alpha f = 0$: set the area above equal to the area below.

④ FTC and functions defined by integrals ★ (repeatedly flagged in reports)

- $d/dx \int_a^x f(t)dt = f(x)$ (write the answer as a function of x); if the upper limit is $g(x)$, multiply by $g'(x)$, e.g. $D(t) = \int_0^{\pi t} \dots \Rightarrow D'(t) = \pi \cdot (\dots)$.
- If the lower limit is a variable: $\int_t^3 f = -\int_3^t f \Rightarrow$ derivative is $-f(t)$.
- Sketch $A(x) = \int_a^x f$: $A(a) = 0$; f changes from positive to negative $\Rightarrow A$ has a maximum, negative to positive $\Rightarrow A$ has a minimum; a turning point of $f \Rightarrow$ a point of inflection of A ; label the value of A at endpoints.
- Determine the sign of $F(x)$: compare the positive and negative areas between 0 and x; the reasoning must refer to area.

⑤ Area

- First find intersection points (often 1 mark on its own) $\rightarrow \int (\text{top} - \text{bottom})dx \rightarrow$ the integral must include limits and dx (2025F-Q5 has 1 mark specifically for notation).
- Cases requiring splitting: finding area between the curve and the x-axis when the curve crosses the x-axis; or when the top/bottom relationship of two curves swaps. Also split when the upper or lower boundary consists of more than one function (2024 report).
- Rectangle subtraction: area enclosed by $y = e^x$, $y = 2$ and the y-axis $= 2 \ln 2 - \int_0^{\ln 2} e^x dx = 2 \ln 2 - 1$.
- Inverse function area: $\ln x$ is below the x-axis when $x < 1$, so a negative sign is needed (2020A-Q11(e)).

⑥ Definite integral equations with parameters: calculate the expression in terms of the parameter = given value, then solve ($\ln b - \ln a = \ln 3 \Rightarrow b = 3a$).

⑦ Riemann sums

- A lower sum takes the smaller function value on each subinterval, an upper sum takes the larger; for an increasing function the left endpoint underestimates and the right endpoint overestimates; for a decreasing function it is the opposite ($1/x$). The explanation is worth 1 mark separately.
- Best estimate $= (L + U)/2$; ways to improve: use more rectangles, or find the function and integrate.

⑧ Context applications: cross-sectional area $\times$ length (width) = volume; total change $= \int$ rate of change; when the water level is unknown, first write the cross-sectional area as a function of h (upper limit contains h), then solve for h ; make sure units are consistent (cm and m).

3 Rectilinear motion

Concept comparison

- Displacement x can be positive or negative; "at rest" $\Leftrightarrow v = 0$; changing direction $\Leftrightarrow v$ changes sign (not counted if v touches 0 and returns to the same sign, e.g. $v = (t - 2)^2$).
- Returning to the starting point $\Leftrightarrow x(t) = x(0)$; passing through the origin $\Leftrightarrow x(t) = 0$ (these are the same only when $x(0) = 0$).

- Time wording: "first appears / initially" is $t = 0$ (2024 report: many substituted $t = 1$); "during the fifth second" is $4 \leq t \leq 5$.

① Speeding up or slowing down ★

- v and a same sign $\Rightarrow$ speeding up; opposite signs $\Rightarrow$ slowing down. Δ Saying only "acceleration is negative" does not earn marks (exact wording from 2024F-Q2 key).
- Δ The 2018A-Q11(c) key treats "heading south and decelerating at 0.5" as $a = -0.5$; for wording like this, write your interpretation and positive direction.

② Working backwards by integrating $a(t)$ or $v(t)$

- Add a constant for each integration; use $v(0)$, $x(0)$ to determine the constants; when there are two position conditions, solve simultaneously for the parameters and constants (2017A-Q20: $p = 12$).
- For trigonometric integrals, divide by the inner coefficient: $\int 2 \sin(t/3 + \pi/6) dt = -6 \cos(t/3 + \pi/6) + C$.

③ Distance travelled and change in displacement ★ guaranteed in exams

- Distance travelled: solve $v = 0$ to find all times when it stops $\rightarrow$ split into intervals $\rightarrow$ put a negative sign in front of the integral on intervals where $v < 0$ (or take the absolute value of each interval) $\rightarrow$ add, with units. The sign is worth 1 mark separately, and writing dt in each integral is also worth 1 mark (2023A-Q8(c)).
- It can also be written as $|x(t_1) - x(0)| + |x(t_2) - x(t_1)| + \dots$ (2024F-Q2(d)).
- Interpreting $\int_a^b v dt$: "the change in displacement from $t = a$ s to $t = b$ s" (the start and end times must be written; 1 mark separately).
- Δ Do not directly use $x(\text{final}) - x(\text{initial})$ as distance travelled.

④ Maximum velocity / zero acceleration

- At an interior extremum, $a = 0$ must hold, but the converse is not necessarily true (in a constant-velocity interval, a is always 0); verify using $v'' < 0$ or a sign table (marks are awarded when the question requires using calculus).
- When finding maximum speed, also compare $|v|$ at the endpoints.

⑤ Interpreting graphs and sign tables: Use the sign of a to explain whether v is increasing or decreasing; 1 mark for each of the two intervals; when given only a sign table for x , v and a : x changes sign $\Rightarrow$ passes through the origin, v changes sign $\Rightarrow$ stops and turns around, v and a opposite signs $\Rightarrow$ slowing down.

4 Exponentials and logarithms

① Logarithm laws and expressing in terms of pronumerals (CF)

- $\log(mn) = \log m + \log n$; $\log(m/n) = \log m - \log n$; $\log m^k = k \log m$; change of base $\log_a b = \log_c b / \log_c a$.
- Write constants as logarithms: $4 = \ln e^4$, $2 = \log_{10} 100$, $9 = \log_{10} 10^9$ (this is the key step in several questions).
- Example: $p = \ln 2$, $r = \ln 5 \Rightarrow \ln 6.25 = \ln(5^2/2^2) = 2r - 2p$; $m = \log_3 6$, $n = \log_6 5 \Rightarrow \log_3 10 = (m - 1) + mn$; $e^{(\ln 2 + \ln 3)} = 6$.

② Solving exponential / logarithmic equations (CF exact values)

- Collect like terms: $4e^{2x} = 81 - 5e^{2x} \Rightarrow e^{2x} = 9 \Rightarrow x = \ln 3$.
- Quadratic in e^x : let $u = e^x$, factorise, reject $u \leq 0$.

- Logarithmic equations: first combine into a single logarithm and then remove the log; Δ check the domain; for $2 \ln x = \dots$, only the positive root can be taken (2025F-Q7 awards marks for only one solution).

③ Logarithmic function graphs and transformations

- $y = m \log_a(x - p) + q$: vertical asymptote $x = p$, passes through $(p + 1, q)$; determine p from the asymptote, then substitute two points to solve for m and q .
- Graphing is worth 3 marks: asymptote (draw it and write its equation), one specific point, correct shape.
- Transformations: $\ln 4x = \ln x + \ln 4$ (translation upwards); $\ln \sqrt{x} = \frac{1}{2} \ln x$ (vertical dilation); $\ln x + 4 = \ln(e^4 x)$ (horizontal dilation); $\log_2(1/(x - 1)) = -\log_2(x - 1)$.
- When reading values from a graph, show the reading process on the graph (2024A-Q15 "show clearly how you used the graph").

④ Growth / decay models $P = P_0 e^{kt}$ ★ high-frequency in CA

- Initial value: $t = 0$; find k from the half-life $\frac{1}{2} = e^{(k \cdot T \frac{1}{2})}$; reaching a certain value: set the model equal to the target value and solve using $\ln$.
- Convert to a time and **round according to the context**: e.g. $4e^{(-0.05t)} = 2.5 \Rightarrow t \approx 9.40 \text{ h} \Rightarrow 9 \text{ am} + 9 \text{ h } 24 \text{ min} = 6:24 \text{ pm}$ (for latest, round down; 6:25 is already too late).
- Rate of change $dP/dt = kP$: if asked for rate of change, give the signed value; if asked for rate of decrease, write it as a positive number and state that it is decreasing.
- Piecewise models: the initial value of the later piece equals the value at the end of the earlier piece; long-term behaviour: as $t \rightarrow \infty$, $e^{(-kt)} \rightarrow 0$.

⑤ Logarithmic scales (decibels, Richter, site rankings)

- $L = k \cdot \log(I/I_0) \Rightarrow I_2/I_1 = 10^{((L_2 - L_1)/k)}$. Δ k differs between questions: decibels $k = 10$ (90 dB is 1000 times 60 dB); sound pressure $k = 20$; Richter $k = 1$; site ranking $k = 2$.
- The difference in readings is not the answer; it must still be converted to a power of 10; to find I_0 : substitute one set of (L, I) values and convert to exponential form.
- If $\log P$ is a straight line against $t \Rightarrow P = 10^A \cdot (10^B)^t$; use the slope to compare growth rates.

- ⑥ **Logarithmic models + differentiation**: determine parameters from known points ($A(0) = 21$, $A(1) = 53 \Rightarrow c = 21$, $b = 64$); $ds/dD = 40.6/D$ decreases as D increases, use it to draw a conclusion; round quantities down; $1 < x < 2$ belongs to "week 2".

5 Discrete random variables and the binomial distribution

① Distribution tables

- For equally likely outcomes, list the sample space (use a table for two dice); when there are unknowns, use $\Sigma P = 1$ and $E(X) = \mu$ to form two equations (1 mark each).
- From the cumulative distribution $P(X \leq x)$: $P(X = k) = F(k) - F(k - 1)$; conditional probability $P(X = 1 \mid X \leq 3) = P(X = 1)/P(X \leq 3)$.
- New variables need cases: number unsold $Y = \max(0, 3 - X)$.

② Expectation, variance, linear transformations

- $E(X) = \Sigma xp$; $\text{Var}(X) = \Sigma x^2 p - \mu^2$; $E(aX + b) = aE(X) + b$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$; $\text{SD}(aX + b) = |a| \cdot \text{SD}$.

- $\Delta \text{Var}(25 - 2X) = 4\text{Var}(X)$; constants do not affect variance.
- Rescale to a new mean and standard deviation: **determine a from the SD first** (take $a > 0$ to preserve the order of rankings), then determine b from the mean. Costs increase by 20% and then a \$1 surcharge is added $\Rightarrow$ new mean $1.2\mu + 1$, new SD 1.2σ .

③ Games / fundraising expectation (examined in 6 of 10 years)

- First define the payoff variable, decide whose perspective you are taking (player or organiser) $\rightarrow$ make a "net payoff + probability" table $\rightarrow E = \Sigma \text{value} \times \text{probability} \rightarrow$ write a conclusion ("on average / in the long run").
- Pricing: set expected profit equal to the target; minimum charge = expected payout (2025A-Q8: $E = 0.8$, \$5 per red ball $\Rightarrow$ at least \$4).

④ Binomial distribution ★

- Write $X \sim \text{Bin}(n, p)$; usually 1 mark for the name and 1 mark for the parameters; state what n and p mean in context.
- $E = np$, $\text{Var} = np(1 - p)$; working backwards from the mean and variance: $np = \frac{1}{2}$, $np(1 - p) = \frac{5}{12} \Rightarrow p = \frac{1}{6}$, $n = 3$.
- Boundaries: more than three $\Rightarrow X \geq 4$; fewer than 2 working in the system $\Leftrightarrow$ at least 4 broken; when the situation changes, change the parameters (removing one $\Rightarrow \text{Bin}(4, p)$).
- Assumptions (2 marks): trials are **mutually independent**; the probability of success **p is the same and constant**; write these in context. "Two outcomes" and "n fixed" are usually given in the question, so writing them may not earn marks.
- Not binomial: the number of trials is not fixed; sampling without replacement makes the trials not independent (improvement: with replacement).
- For very small probabilities, keep more decimal places (2019A-Q18(c) about 0.00003; the key requires at least 5 decimal places).

- ⑤ **Normal $\rightarrow$ binomial**: first use the normal distribution to find the probability p for a single item (keep enough decimal places) $\rightarrow Y \sim \text{Bin}(n, p)$ (write "binomial" and the parameters) $\rightarrow$ find $P(Y \geq k)$.

- ⑥ **Paths and counting**: translate the event into a number of successes. Example: returning to the origin $\Leftrightarrow$ number of forward steps = number of backward steps (impossible in an odd number of steps, probability 0, give a reason); moving forward $\geq 20 \text{ cm}$ (5 cm per step) $\Leftrightarrow$ at least 7 forward steps out of 10.

6 Continuous Random Variables and the Normal Distribution

① Histogram $\rightarrow$ probability (CF)

- Add relative frequencies; for a frequency histogram, divide by the total first.
- When an endpoint falls inside a bar, interpolate proportionally (provided the question states 'uniform within each class'); $E \approx \Sigma \text{class midpoint} \times \text{probability}$; distinguish clearly between cumulative probability tables and individual probabilities.

② pdf: the four essentials

- Total area = 1 (write this step whether finding k or proving k); $P(a < X < b) = \int_a^b f$; $E = \int x f$; $\text{Var} = \int x^2 f - \mu^2$

or $\int (x - \mu)^2 f$; median $\int^M f = 0.5$.

- Given cdf $F(x)$: $P(a < X \leq b) = F(b) - F(a)$; pdf $= F'(x)$.
- In CF papers, use triangle and trapezium areas for a pdf made of straight-line segments; integrate piecewise pdfs separately.
- Example: $f = 1/(x \ln 5)$, $1 \leq x \leq 5 \Rightarrow E = 4/\ln 5$, median $\sqrt{5}$.
- Convert the expectation to context units or time (3.8 h after noon $\Rightarrow$ 3:48 pm); the units of variance are squared units.

③ Uniform distribution

- $f = 1/(b - a)$, state the domain $a \leq x \leq b$ and 0 otherwise; $E = (a + b)/2$.
- Δ For variance, answer by writing the integral (the key awards 1 mark each for 'pdf and domain' and 'integral expression'); $(b - a)^2/12$ is not on the formula sheet, so use it only to check. Example: $E = 6$, maximum 9 $\Rightarrow U(3, 9)$, $\text{Var} = \int_3^9 (x - 6)^2/6 dx = 3$.
- Conditional probability: $P(X < -0.35 \mid X < 0) = 0.15/0.5$; $X^2 < 0.09 \Leftrightarrow -0.3 < X < 0.3$.

④ Normal distribution

- First write the probability statement (in most keys this is 1 mark on its own), then calculate; inverse quantile: $P(X > m) = 0.0005 \Rightarrow \text{invNorm}(0.9995)$.
- Conditional probability: $P(X < 75 \mid X > 67) = P(67 < X < 75)/P(X > 67)$.
- Change of units $Y = aX + b$: $\mu_Y = a\mu + b$, $\sigma_Y = |a|\sigma$, $\text{Var}(Y) = a^2\text{Var}(X)$; it may also be reversed: find a, b from μ_Y, σ_Y .

⑤ Find μ, σ from two tail probabilities ★

- Find two z-values (1 mark) $\rightarrow$ set up equations $\rightarrow$ solve. Example: $5\% < 55$, $13\% > 100 \Rightarrow z = -1.6449, 1.1264 \Rightarrow \sigma \approx 16.24, \mu \approx 81.71$.
- $P(X \leq 150) = 0.0228$, $P(X \geq 165) = 0.1587$ correspond exactly to $z = -2, z = 1 \Rightarrow \sigma = 5, \mu = 160$. When only the mean is known, one z is enough.

⑥ 68–95–99.7 (CF): write values as $\mu \pm k\sigma$; the flatter and wider the curve, the larger σ is; the probability for an interval of width σ is at most about 0.38.

⑦ Reasons a normal model is not appropriate: histogram is asymmetric or skewed; variable is bounded; model gives impossible values (e.g. $P(\text{weight} < 0) > 0$).

7 Sample Proportions and Confidence Intervals

① Distribution of $\hat{p}$

- Write 'approximately normal (n sufficiently large) + mean p + variance $p(1 - p)/n$ ' for 3 marks in total, then find the probability.
- Distribution is asymmetric (e.g. $n = 36, p = 0.05 \Rightarrow$ the normal approximation cannot be used).
- As n increases $\Rightarrow$ standard deviation of $\hat{p}$ decreases and the distribution becomes more concentrated $\Rightarrow$ probabilities such as $P(\hat{p} \leq 0.03)$ become larger (1 mark for each of the two points).

② Calculating confidence intervals

- $\hat{p} \pm z\sqrt{\hat{p}(1 - \hat{p})/n}$, $z = 1.645$ (90%) / 1.960 (95%) / 2.576 (99%); round to the required number of decimal places.
- Work backwards from an interval: $\hat{p}$ = midpoint, E = half-width; given the interval and n , work back to the confidence level: $z = E/SE$, then look up the percentage.

③ Sample size Δ Most error-prone

- For the minimum sample size (minimum / guarantee / ensure margin at most), always use $\hat{p} = 0.5$, even if $\hat{p}$ has already been calculated earlier (2023A-Q7(c): using 0.5 gives 2401, substituting 0.38 gives 2263, losing marks). Only substitute a particular sample proportion if the question explicitly says to use it.
- 'At least' $\Rightarrow$ round up ($259.2 \Rightarrow 260$); working backwards from an interval to the original $n \Rightarrow$ round to the nearest integer ($350.1 \Rightarrow 350$).
- Proportional relationships: error halved $\Rightarrow n \times 4$; width becomes $1/3 \Rightarrow n \times 9$.

④ Using an interval to test a claim ★ Appears every year

- Two sentences: 'The claimed value ... is (not) within the CI ...' + 'so there is sufficient / insufficient evidence to conclude ...'. Δ You must take a position; you cannot avoid it by saying 'cannot be 100% certain' (2023 report).
- Two intervals overlap $\Rightarrow$ there is insufficient evidence to say the two are different.
- If asked "does it **prove** ... / was a mistake made" $\Rightarrow$ answer **No**: not all intervals contain p , and you don't know whether this interval is one that did not contain p (2025 report).

⑤ Factors affecting interval width (1 mark for factor + 1 mark for effect): increasing n makes it narrower; increasing the confidence level makes it wider; $\hat{p}$ closer to 0.5 makes it wider.

⑥ Sampling bias

- How to write it: identify the source + explain which type of people are more or less likely to be selected + link to the context; time and place are two separate sources and must be written separately (2022 report).
- Common sources: location (only outside a shopping centre or library), time (weekday lunchtime), voluntary response, leading questions, sampling frame (mobile phones only), convenience sampling (the first 200 printed books).
- Improvement: random sampling across the whole population (the whole week, all printing presses).

⑦ Other common question types

- The number of k 90% intervals containing $p \sim \text{Bin}(k, 0.9)$; '3 intervals do not contain p ' $\Leftrightarrow X = k - 3$.
- Assumptions of the binomial distribution: state the assumption $\rightarrow$ decide whether it is reasonable in the context $\rightarrow$ give a reason.